1 Introduction

This project focuses on the design, development, and implementation of an interactive engineering tool that integrates electrical and systems control systems. The objective is to develop a functional and reliable system that allows users to interact with various electrical components and seamlessly import or create one-line diagrams within a specialized component GUI. The software will validate system topology, map zones of protection, and perform critical momentary DC battery load calculations. By identifying worst-case scenarios, the tool provides GFT engineers with actionable insights to ensure maximum substation reliability and safety.

The project provides an opportunity to apply theoretical concepts learned through coursework to the creation of a practical engineering application. Key areas of focus include system design, component integration, programming, testing, and troubleshooting. The completed system is intended to demonstrate the application of engineering methods to solve a real-world problem.

The GUI allows users to construct protection system schematics using a Drag-and-Drop Approach, identifying and visualizing zones of protection, performing momentary current calculations, and accessing a library of electrical components. In addition, the tool provides the ability to import an existing PDF diagram and use it as a background reference while recreating or analyzing the system. These features are intended to make the analysis process more clean, organized, interactive, and easier to understand.

The tool is divided into several primary sections: the Protection Zone Outliner, Momentary Calculation Tool, Component Library, Sandbox, and PDF Importer/Exporter function.

1.1 Project Objectives

2 Tool Overview

2.1 Graphical User Interface

The graphical user interface (GUI) serves as the central workspace for the protection analysis tool. The GUI organizes the different functions into accessible sections, allowing the user to move between diagrams, component selection, protection zone analysis, and electrical calculations without needing to leave the

application. The interface is designed around an interactive workspace where components can be placed, connected, and modified as the protection system is developed.

2.2 Component Library

The component library can also be expanded through the Create a component functionality button. This allows a user to create a custom component when a required component is not already included in the library. The ability to create customized components makes the tool more flexible and allows it to accommodate different protection-system configurations and project requirements. The Component Library provides the users with a collection of electrical and protection system components that industries use that can be used to construct one-line diagrams. The Component Library allows for both commonly used and custom created components to be selected and placed into a workspace for quicker access and efficiency, instead of needing the user to manually draw each component into the schematics.

2.3 Protection Zone Outliner

The Zone Outliner will be used to organize and visualize the different areas of Zones of Protection found within the electrical schematics. Within the sandbox, once the components and circuit connections are established, the Zone Outliner would create a display that would identify the protection zones. The purpose of having this feature is make it easier in relationships between the schematic components and their associated protection zones to be easier to understand. Separating the electrical system into individual zones, would allow easier identification among it's users for which components are included within a particular zone and how evaluating the different parts of the system is protected.

2.4 Momentary Calculation

The Momentary Calculation Tool provides a section that is solely focused on performing calculations associated with the momentary electrical load of the protection system for the given One-Line Diagrams. The tool is intended to reduce the need for the user to perform these calculations separately and manually. By providing the calculation functionality within the same GUI used to develop the protection diagram, the user can connect the calculated results to

the components and protection zones that are being analyzed in the GUI. This helps maintain a workflow that is concentrated and allows the electrical analysis to be performed alongside the development of the system diagrams in pending projects.

2.5 PDF

After importing the PDF, the diagram can be displayed in the background of the Sandbox while the user recreates with additional component edits/changes or analyzes the system using the available components. The background image as well as the grid can be toggled on or off, allowing the users to switch between viewing the original diagram and viewing of the newly created components within the sandbox.

2.6 Sandbox

The Sandbox is the main workspace where users can create, edit, and analyze their protection system diagrams.